RESEARCH NAVIGATOR

Full Product, UX, Data, Crawler & Implementation Specification

Working title - brand name to be finalised

Product thesis
A free, open-source research discovery and planning platform that helps researchers move from a research question to a defensible dataset plan. It indexes datasets rather than re-hosting them, lets users collect datasets into a project basket, checks metadata and compatibility using deterministic rules, and exports a structured research data report with official source links, licences, citations and gaps.

Principle	Decision
V1 intelligence	No paid AI dependency. Deterministic search, rules, filters, metadata scoring, templates and compatibility logic.
Data strategy	Index metadata and official links. Do not mirror third-party datasets unless the licence explicitly permits it and hosting is deliberately enabled.
Audience	Researchers, PhD students, academics, consultants, students, planners, analysts, public-sector teams and interdisciplinary project groups.
Initial domain	Built environment, architecture, energy, climate, GIS and urban research; architecture must allow later expansion to economics, health, transport, social science and other domains.
Core differentiator	Not just “find a dataset”. Build a project-ready dataset portfolio and document why each dataset belongs in the study.

1. Executive product definition

Research Navigator is a dataset discovery, research-planning and evidence-packaging platform. It behaves more like a research guide than a file repository. A user starts from a question, topic, method or known dataset; discovers relevant official datasets; places selected datasets into a Project Basket; reviews licence, coverage, variables, spatial/temporal resolution and compatibility; and exports a project report containing a dataset inventory, source links, citations, coverage matrix, identified gaps and next-step checklist.

1.1 The problem it solves

- **Fragmented discovery.** Researchers repeatedly search government portals, institutional repositories, APIs, GIS catalogues and project websites for datasets that are difficult to compare.
- **Metadata inconsistency.** The same practical questions recur: What geography does this cover? What years? What variables? What licence? Is there an API? Can it be combined with another source?
- **Research planning friction.** Dataset selection is rarely documented in one place, making methods less reproducible and forcing researchers to rebuild dataset inventories manually.
- **Licensing risk.** Re-hosting data can be legally or operationally complicated. The platform avoids this by defaulting to metadata indexing and linking back to authoritative sources.
- **Cross-dataset compatibility.** A “good” dataset in isolation may be useless for a project if its geography, time period, identifiers or resolution do not align with other selected sources.

1.2 Product promise

One-line promise
Ask a research question, find the right datasets, assemble them into a project, understand how they fit together, and leave with a citable research data plan.

1.3 What the product is not

- Not a replacement for Kaggle or Zenodo.
- Not an unlicensed data mirror.
- Not an automatic statistical analysis system in V1.
- Not a black-box AI research assistant in V1.
- Not restricted to EPW, EnergyPlus or built-environment data in its long-term architecture.

2. Target users and jobs-to-be-done

User	Typical situation	Primary outcome
PhD / MSc researcher	Starting literature, fieldwork or modelling and unsure which datasets exist.	Create a defensible dataset plan and export it for supervisors / methods chapter.
Academic / PI	Designing a proposal or multi-partner study.	Rapidly scope data availability, licences, gaps and dependencies.
Consultant / analyst	Needs authoritative sources for a client study.	Find reliable data quickly and document provenance.
GIS / spatial researcher	Combines boundaries, demographics, buildings, climate and transport data.	Check spatial compatibility, CRS, resolution and geography.
Building performance researcher	Needs weather, EPC, building stock, morphology and simulation context.	Assemble a complete modelling evidence package.
Student / beginner	Does not know what datasets are usually required.	Use curated Research Kits and guided templates.

Dataset publisher / institution	Wants a dataset to be discoverable without duplicating storage.	Submit metadata and drive users to the official source.

2.1 Core jobs-to-be-done

1. Find authoritative datasets for a topic or research question.
2. Compare similar datasets before choosing one.
3. Understand practical constraints before downloading.
4. Build a multi-dataset research project basket.
5. See whether selected datasets can realistically work together.
6. Generate a shareable dataset plan with persistent project state.
7. Cite datasets correctly and return to authoritative sources.
8. Identify missing data and alternatives.
9. Save curated dataset combinations as reusable project templates.

3. Core user journeys

3.1 Journey A - Start from a research question

10. User enters a research question, e.g. “How does urban form influence summertime overheating risk in UK dwellings?”
11. The system extracts no AI meaning in V1; instead it offers structured topic chips, domain filters, method filters and user-selected keywords.
12. Search results show datasets ranked by relevance, metadata completeness, authority and filter fit.
13. User opens dataset cards and compares coverage, variables, granularity, access, licence and update frequency.
14. User clicks Add to Project Basket.
15. Basket groups datasets by research role: context, exposure, outcome, geometry, weather, validation, socio-demographics, boundaries, etc.
16. Compatibility engine flags likely mismatches and missing companion data.
17. User exports the Research Data Plan as PDF/HTML/CSV/JSON.

3.2 Journey B - Start from a known dataset

18. User searches “EPC England and Wales”.
19. Dataset Detail page shows official publisher, licence, variables, dates, spatial unit, download/API links, documentation and related datasets.
20. The user adds it to a project.
21. The platform shows common companion datasets such as building footprints, address identifiers, boundary datasets or weather sources based on deterministic relationship tables.
22. The user generates a project pack.

3.3 Journey C - Start from a Research Kit

23. User opens Research Kits.
24. Chooses a template such as “UK Residential Overheating Study”, “Urban Heat Island Mapping”, “Housing Energy Vulnerability”, “Walkability & Accessibility”, or “Building Stock Retrofit Analysis”.
25. A pre-populated basket appears with required, recommended and optional data roles.
26. User swaps datasets according to geography or year.

27. The system re-evaluates compatibility and completeness.
28. User saves the kit as their own project.

4. Inputs and outputs

4.1 User inputs

Input	Examples	Required?
Research question	Free text project question.	Optional
Aims / objectives	One or more project objectives.	Optional
Keywords	overheating, housing, Leeds, urban heat island	Optional
Domain	Built environment, economics, health, transport, etc.	Recommended
Geography	UK, England, Leeds, neighbourhood, global	Recommended
Time period	2010-2025, current year, historic, future	Optional
Methods	GIS, regression, simulation, survey, ML, time-series	Optional
Target variables	temperature, EPC rating, income, building height	Optional
Dataset selections	Items added to Project Basket.	Core
Own dataset metadata	User may add a private/local dataset record to the project without uploading the actual data.	Optional

4.2 Platform outputs

- **Dataset discovery results.** Searchable, filterable catalogue of indexed dataset metadata.
- **Dataset detail pages.** Human-readable technical summaries and authoritative access links.
- **Project Basket.** A structured, editable collection of datasets for one research project.
- **Compatibility report.** Deterministic checks on geography, time, resolution, identifiers, format and licence.
- **Coverage matrix.** What the project covers versus what is still missing.
- **Research Data Plan.** Exportable report with research question, aims, selected datasets, rationale, provenance, licence, access, citations, warnings and gaps.
- **Bibliography block.** Dataset citations in APA/Harvard/BibTeX/RIS-like structured export where source metadata supports it.
- **Link bundle.** All official download/documentation/API links in one place.
- **Machine-readable manifest.** CSV/JSON containing dataset metadata and source URLs for reproducibility.

5. Functional feature map

Module	V1 capability	Priority
Global Search	Full-text metadata search + filters + sort.	P0
Dataset Catalogue	Cards, list view, map-aware filters, metadata quality indicators.	P0
Dataset Detail	Technical metadata, provenance, licence, access links, related datasets.	P0
Project Basket	Add/remove/group/annotate datasets;	P0

	persistent state.	
Project Workspace	Question, aims, methods, geography, period, notes and datasets.	P0
Compatibility Engine	Rule-based pairwise and project-level checks.	P0
Research Report Generator	HTML/PDF/CSV/JSON project pack.	P0
Research Kits	Curated project templates.	P1
Crawler / Harvester	Fetch public metadata from APIs, sitemaps, CKAN/DCAT/catalogues and approved pages.	P1
User Submissions	Submit a dataset URL and metadata for moderation.	P1
Saved Searches	Bookmark filters and optional email/RSS later.	P2
Dataset Watch	Track dataset metadata/update-date changes.	P2
Publisher Dashboard	Manage claimed catalogue entries.	P2
Optional AI Layer	Future only; never required for core operation.	P3

6. Complete website information architecture

Page	Contents
Home	Search-first landing page; research-question box; topic chips; featured datasets; popular research kits; “Browse by domain”; “Browse by geography”; statistics; open-source/GitHub CTA.
Explore Datasets	Faceted catalogue with search, filter sidebar, sort, card/list toggle, selected-filter chips, result count, pagination/infinite scroll, Add to Basket on every card.
Dataset Detail	Title, publisher, authority badge, description, coverage, variables, spatial/temporal resolution, identifiers, formats, licence, update schedule, access method, official links, documentation, source status, metadata completeness, related datasets, compatible-with list, citation, changelog, report issue.
Compare Datasets	Side-by-side comparison of up to 4 records across coverage, variables, resolution, years, formats, licence, API, access friction, update frequency and metadata quality.
Project Basket	Persistent side drawer + full page; dataset grouping; quantity not relevant; notes; role assignment; duplicate warning; compatibility summary; “Create Project” / “Generate Report”.
Project Workspace	Research question, aims, geography, date range, methods, variables, dataset table, compatibility panel, missing-data checklist, notes, collaborators later, export actions.
Research Kits	Template library with cards showing research purpose, typical methods, dataset roles, geography support, difficulty, required/recommended/optional datasets.
Research Kit Detail	Purpose, expected outputs, workflow, data roles, default datasets, alternatives, known caveats, “Use this kit”.
Coverage Map	Optional map-first discovery page; filter datasets by geographic coverage and spatial scale.
Publish / Submit Dataset	URL-first submission, metadata form, licence selection, publisher identity, moderation status.
Publishers	Browse authoritative publishers, dataset counts, domains,

	access types and publisher profile pages.
My Projects	Saved project cards, last edited, completeness, dataset count, export, duplicate, archive.
Saved / Favourites	Saved datasets and saved searches.
About & Methodology	Explain indexing model, rankings, compatibility rules, licence policy and open-source governance.
API / Developer	Public metadata API docs, rate limits, schema, examples.
Admin / Curator	Queue, crawler health, source registry, metadata conflicts, licence flags, duplicate merge, broken-link checks, audit log.

7. Dataset catalogue UX specification

7.1 Dataset card

- Dataset title and short one-line description.
- Publisher / institution.
- Authority badge: Official / Academic / Community / Unknown.
- Domain tags and variable tags.
- Geographic coverage chip.
- Temporal coverage chip.
- Spatial resolution or unit if applicable.
- Access type: Direct Download / API / Portal / Registration / Request.
- Licence chip.
- Last updated date.
- Metadata completeness score.
- Status: Active / Deprecated / Archived / Link issue.
- Primary CTA: Add to Basket.
- Secondary CTA: View Dataset.

7.2 Dataset metadata fields

Group	Fields
identity	id, slug, title, alternate_title, short_description, full_description
publisher	publisher_id, publisher_name, publisher_type, authoritative_flag
classification	domains, topics, keywords, research_roles, methods_supported
geography	countries, admin_areas, bbox, geometry, geographic_description, spatial_unit
time	start_date, end_date, temporal_resolution, update_frequency, latest_release_date
variables	variable_name, label, unit, datatype, description, vocabulary_id
spatial	spatial_resolution_value, spatial_resolution_unit, CRS, geometry_type
access	landing_page_url, download_url, api_url, documentation_url, registration_required, access_notes
format	CSV, GeoJSON, Shapefile, GPKG, Parquet, NetCDF, EPW, XLSX, API, etc.
licence	licence_id, licence_name, licence_url, reuse_allowed,

	commercial_reuse, attribution_required
provenance	source_catalogue, source_record_id, source_first_seen, source_last_checked
quality	metadata_score, broken_link_state, freshness_state, curator_status
citation	authors/organisation, publication_year, version, DOI/identifier, suggested_citation
relationships	replaces, replaced_by, derived_from, companion_to, alternative_to, overlaps_with

8. Search, discovery and ranking

8.1 Search facets

- Domain / discipline
- Topic
- Geography
- Temporal coverage
- Spatial scale
- Temporal resolution
- Data format
- Access method
- Licence
- Publisher
- Official/academic/community source
- Variables
- Research method
- Update frequency
- Free/open access only
- API available
- Registration required
- Metadata completeness
- Last updated

8.2 Deterministic ranking formula

V1 ranking can combine indexed text relevance with explicit, explainable boosts. A recommended starting score is:

Ranking score
0.55 × text relevance + 0.15 × filter fit + 0.10 × authoritative-source score + 0.08 × metadata completeness + 0.07 × freshness + 0.05 × link-health score. All components are normalised to [0,1].

Ranking must never silently imply scientific quality. The UI should label it as “search relevance”, not “best dataset”.

9. Project Basket and research workspace

9.1 Basket behaviour

- Basket persists for anonymous users in local storage and synchronises to account after sign-in.
- A dataset can belong to one or more project roles.
- Users can write a rationale note for every dataset.
- The basket displays warnings immediately without blocking selection.
- Users can compare items, replace an item with an alternative, or mark a dataset “required”.
- Basket can be converted to a named project with one click.

9.2 Dataset research roles

Role	Examples
Study boundary	Administrative boundary, study-area polygon, postcode geography
Population / demographics	Census, deprivation, household composition
Built environment	EPC, building footprints, heights, land use, LCZ
Climate / exposure	Weather, temperature, rainfall, flood, air quality
Outcome	Overheating, energy use, health outcome, economic indicator
Validation / reference	Ground station, survey benchmark, official statistic
Mobility / accessibility	Road, transit, walking network
Economic / policy	Income, prices, grants, policy boundaries
Lookup / identifiers	UPRN, postcode lookup, code lists

10. Compatibility engine - no AI required

Compatibility is a rule engine, not a guarantee that two datasets are scientifically appropriate. It provides explicit warnings that a researcher can inspect.

Check	Logic	User-facing result
Geographic overlap	Do dataset bounding boxes / administrative areas intersect the project geography?	Block only when no overlap.
Spatial granularity	Are units compatible or can one be aggregated / spatially joined?	Warn on large mismatch.
CRS	Are spatial coordinate systems known and transformable?	Warn when unknown.
Temporal overlap	Do periods overlap the project target period and each other?	Warn / fail if no overlap.
Temporal resolution	Hourly vs annual; daily vs monthly.	Warn about aggregation implications.
Join identifiers	Shared identifiers such as UPRN, OA code, postcode, coordinates?	Recommend spatial join if absent.
Licence	Can selected datasets legally be combined/published under intended reuse?	High-priority warning.
Access	Registration or restricted-access dependencies.	Warn before project export.
Format	Can the formats be handled in typical GIS/Python/R workflows?	Informational.
Freshness	Is one dataset substantially older than others or the target period?	Warn.

Unit conventions	Known variable units conflict?	Warn where variable-level metadata exists.

10.1 Project completeness engine

Each Research Kit defines required data roles. Completeness is calculated as role coverage, not number of datasets. Example: an Urban Heat Island kit may require study boundary, urban morphology, temperature exposure, rural/reference climate and validation metadata. If 4 of 5 required roles are filled, completeness = 80%. Optional roles do not reduce the score.

11. Research Data Plan / report output

11.1 Report sections

29. Cover and project metadata.
30. Research question, aims and scope.
31. Dataset inventory table.
32. Why each dataset is included.
33. Authoritative source and direct access/documentation links.
34. Licence and reuse summary.
35. Spatial coverage matrix.
36. Temporal coverage matrix.
37. Variable / research-role coverage matrix.
38. Compatibility warnings and required transformations.
39. Missing-data and unresolved-dependency checklist.
40. Alternative datasets.
41. Suggested acquisition order.
42. Dataset citations and identifiers.
43. Reproducibility manifest.
44. Appendix: raw metadata snapshot and link-check timestamps.

11.2 Export formats

- PDF - human-readable project pack.
- HTML - shareable static report.
- CSV - dataset inventory.
- JSON - reproducible machine-readable manifest.
- BibTeX/RIS - dataset references where metadata permits.
- Markdown - methods notebook / GitHub-friendly project plan.

12. Research Kits

Research Kits are curated, versioned templates. They are one of the strongest differentiators because they turn the catalogue into a research planning tool without AI.

Kit	Core data roles	Typical methods
Urban Heat Island Mapping	Boundary; temperature; reference weather; morphology; land cover;	GIS + time-series

	population	
UK Residential Overheating	Weather/DSY; building stock/EPC; geometry/archetype; occupancy assumptions; climate context	Simulation + GIS
Building Stock Retrofit	EPC; building geometry; construction age; energy prices; retrofit cost; carbon factors	Stock modelling
Housing Energy Vulnerability	EPC; fuel poverty; household income; tenure; demographics; weather	GIS + social analysis
Walkability & Accessibility	Street network; destinations; population; public transport; boundary	Network analysis
Flood Exposure & Housing	Flood hazard; building footprints; addresses; vulnerability; elevation	GIS risk analysis
Economic Place Analysis	Businesses; employment; income; boundaries; mobility; property / rents	Spatial economics

13. Dataset crawler and harvesting architecture

13.1 Source priority

45. Structured APIs and catalogue feeds first: CKAN, DCAT, Socrata, ArcGIS REST, OGC catalogues.
46. XML/JSON sitemaps and machine-readable metadata.
47. Repository APIs: Zenodo, Figshare, institutional repositories where permitted.
48. Publisher-specific HTML extraction only when terms permit and no structured interface exists.
49. Manual curator entry and community submission as a fallback.

13.2 Harvester pipeline

50. Source Registry triggers scheduled connector job.
51. Fetch metadata only; respect robots.txt, terms, rate limits and API quotas.
52. Parse source record into canonical schema.
53. Normalise licences, dates, geographies, formats and publisher names.
54. Generate stable source fingerprint to detect duplicates and changes.
55. Run metadata validation.
56. Run URL health checks.
57. Upsert candidate record into staging.
58. Resolve duplicate / conflict rules.
59. Publish automatically only for trusted sources; otherwise moderation queue.
60. Store source snapshot hash and last-checked timestamp.

13.3 Crawler safeguards

- Never bypass authentication, paywalls or access restrictions.
- Do not download bulk dataset content unless explicitly configured and licensed.
- Per-domain rate limits and exponential backoff.
- User-agent identifies the project and provides contact URL.
- Robots and terms compliance.
- Source-level kill switch.
- Immutable audit log of automated changes.

- Broken-link checker should use HEAD where reliable, then minimal GET fallback.
- Crawler must be idempotent.

14. Licensing and legal-by-design model

Default policy
Index metadata and link to the publisher. Do not claim ownership and do not re-distribute source data by default.

- Store the source-declared licence name and URL, plus a normalised licence family.
- Display “Licence not verified” when source metadata is absent or ambiguous.
- Do not infer that “publicly downloadable” means “openly reusable”.
- Record whether commercial reuse, redistribution and derivative use appear permitted only when the licence explicitly states this.
- Every dataset page prominently links to the publisher and source terms.
- Allow publishers to request correction or removal of metadata.
- Use cached metadata responsibly and retain provenance.
- Create a takedown/contact process and source correction workflow.
- For user-submitted records, require submitter acknowledgement that they are submitting metadata/link information, not transferring ownership of the dataset.

15. Database schema

Table	Core fields
users	id, email, display_name, role, created_at, preferences_json
publishers	id, name, slug, type, website, country, verified, description
datasets	id, slug, title, short_description, full_description, publisher_id, authority_level, status, first_seen_at, last_seen_at
dataset_versions	id, dataset_id, version_label, release_date, metadata_snapshot_json, source_record_id
dataset_access	id, dataset_id, landing_url, download_url, api_url, docs_url, access_type, registration_required
dataset_licences	id, dataset_id, licence_id, source_text, verified_state
licences	id, SPDX_or_internal_code, name, url, open_flag, commercial_flag, attribution_flag, redistribution_flag
dataset_geographies	id, dataset_id, country_code, admin_level, admin_code, bbox, geom, geography_text
dataset_temporal	id, dataset_id, start_date, end_date, temporal_resolution, update_frequency
dataset_formats	id, dataset_id, format_code, mime_type
variables	id, canonical_name, label, description, unit, vocabulary_source
dataset_variables	dataset_id, variable_id, source_name, unit_override, datatype
tags	id, slug, label, tag_type
dataset_tags	dataset_id, tag_id
relationships	id, source_dataset_id, target_dataset_id, relation_type, evidence, curated_flag
sources	id, name, base_url, source_type, trust_level, crawl_policy, active

harvest_records	id, source_id, external_id, dataset_id, raw_metadata_json, fingerprint, fetched_at
link_checks	id, dataset_id, url, status_code, checked_at, result
projects	id, user_id, title, research_question, aims_text, geography_json, start_year, end_year, methods_json, visibility
project_datasets	project_id, dataset_id, role, required_flag, rationale, notes, sort_order
research_kits	id, slug, title, description, domain, version, visibility
kit_roles	id, kit_id, role_name, requirement_level, description
kit_dataset_options	kit_role_id, dataset_id, priority, geography_rules_json
compatibility_rules	id, rule_code, scope, severity, description, config_json, active
compatibility_results	project_id, rule_id, dataset_a_id, dataset_b_id, severity, message, details_json, evaluated_at
submissions	id, user_id, source_url, proposed_metadata_json, status, moderator_notes
audit_log	id, actor_type, actor_id, action, entity_type, entity_id, before_json, after_json, created_at

16. API design

Endpoint	Purpose
GET /api/v1/datasets	Search/list datasets with facets.
GET /api/v1/datasets/{slug}	Dataset detail.
GET /api/v1/datasets/{id}/related	Curated/deterministic related datasets.
GET /api/v1/publishers	Publisher list.
GET /api/v1/tags	Controlled vocabularies.
POST /api/v1/projects	Create project.
GET /api/v1/projects/{id}	Project workspace data.
POST /api/v1/projects/{id}/datasets	Add dataset to project.
DELETE /api/v1/projects/{id}/datasets/{datasetId}	Remove dataset.
POST /api/v1/projects/{id}/evaluate	Run compatibility/completeness rules.
GET /api/v1/projects/{id}/report	Render report data model.
POST /api/v1/exports	Generate PDF/HTML/CSV/JSON export.
GET /api/v1/kits	List research kits.
POST /api/v1/kits/{id}/instantiate	Create project from kit.
POST /api/v1/submissions	Submit dataset metadata/URL.
GET /api/v1/geographies/search	Geography autocomplete.

17. Technical architecture

17.1 Recommended stack

Layer	Recommendation	Reason
Frontend	Next.js + TypeScript + React	SEO, fast catalogue pages, strong app UX, open-source ecosystem.
UI	Tailwind CSS + accessible component primitives	Fast consistent design and responsive behaviour.
Backend	FastAPI (Python) or Next.js API for thin endpoints	Python is useful for metadata rules, geospatial processing and crawlers.
Database	PostgreSQL + PostGIS	Relational metadata + spatial

		coverage.
Search	PostgreSQL FTS initially; Meilisearch/OpenSearch later	Avoid unnecessary infrastructure in MVP.
Queue	Redis + Celery/RQ/Arq	Harvesting, link checks and export jobs.
Crawler	Python httpx + source adapters + BeautifulSoup only when needed	Deterministic, maintainable connectors.
Object storage	S3-compatible storage	Reports, metadata snapshots; not source datasets by default.
Auth	Auth.js / Clerk alternative / self-hosted OIDC	Accounts for projects; anonymous browsing stays open.
Deployment	Docker Compose for dev; managed Postgres + container host in prod	Reproducible open-source deployment.
Observability	OpenTelemetry + Sentry-compatible or self-hosted equivalent	Crawler and API reliability.

17.2 Service boundaries

- Web app/API service.
- Metadata search service.
- Harvester worker.
- Link-health worker.
- Compatibility engine.
- Export/report service.
- Admin/curation service.
- Optional notification service later.

18. UX and visual design system

The product should feel like a serious research instrument, not a generic dashboard. The interface must optimise scanning, comparison, provenance and confidence.

Element	Specification
Layout	Wide desktop research workspace with 12-column grid; strong responsive collapse for mobile.
Navigation	Top nav: Explore, Research Kits, Projects, Publishers, About; persistent Basket icon with count.
Search	Large global search field; keyboard shortcut “/”; query persists across navigation.
Dataset cards	Dense but readable; 2-3 line descriptions; metadata chips; authority/licence visible without opening detail.
Colour	Neutral research palette with colour reserved for semantic states: success, warning, restriction, official source, missing metadata.
Typography	High-legibility sans serif; tabular numerals for dates/resolution.
Tables	Sticky headers, column chooser, responsive horizontal scroll, export button.
Map	Use only where geography adds value; never make map the only discovery route.
Accessibility	WCAG 2.2 AA target; full keyboard use, visible focus, semantic headings, contrast-compliant states.
Dark mode	Optional P2, not required for MVP.

18.1 Search result card information hierarchy

61. Title + source authority.
62. One-sentence purpose.
63. Geography + years + resolution.
64. Variables / topic tags.
65. Licence + access method.
66. Metadata freshness / completeness.
67. Add to Basket CTA.

19. Trust, quality and provenance

- Every displayed fact should map to source metadata or a curator field.
- Show “last checked” timestamps.
- Differentiate Publisher-provided, Harvested, Community-submitted and Curator-enriched metadata.
- Expose the platform methodology for metadata completeness and relevance ranking.
- Keep an audit log of manual curation.
- Do not label datasets “high quality” based solely on metadata completeness.
- Dataset status warnings: deprecated, superseded, unavailable, access restricted, licence unclear, stale link.
- Allow users to report metadata errors and broken links.

19.1 Metadata completeness score

Suggested scoring
Title 5%, publisher 10%, description 10%, geography 10%, temporal coverage 10%, licence 15%, access link 10%, documentation 5%, variables 10%, format 5%, update information 5%, citation/identifier 5%. Missing licence must be visually emphasised even if the total score remains moderate.

20. Initial built-environment catalogue scope

The first release should deliberately seed one domain deeply before broadening. This gives the platform a real use case and allows the compatibility engine to be tested against known research workflows.

Category	Example dataset families to index
Building stock	EPC, national/building stock databases, building footprints, addresses/UPRN where legally accessible.
Weather & climate	EPW/TRY/DSY catalogues, Met Office/MIDAS, ERA5, UK climate projections, gridded climate products.
Urban morphology	Building height, land cover, Local Climate Zones, tree canopy, surface materials where available.
Energy	Subnational consumption, smart-meter/open aggregated data, carbon factors, energy prices.
Planning & land	Local plans, land use, planning applications, brownfield registers, constraints.
Population & vulnerability	Census, deprivation, fuel poverty, health/vulnerability aggregates.
Transport	Roads, public transport, active travel, accessibility datasets.
Hazards	Flood, heat, air quality, noise and other environmental exposure layers.

21. Source registry strategy

Do not hard-code “the internet”. Maintain a Source Registry. Every source adapter has an owner, crawl method, cadence, legal notes and trust level.

Source type	Adapter strategy	Typical cadence
CKAN portal	Native API harvester	Daily/weekly
ArcGIS Hub / REST	Item search + service metadata	Weekly
DCAT catalogue	RDF/JSON-LD feed parser	Weekly
Socrata	Catalogue API	Weekly
Zenodo / Figshare	Repository API	Weekly
Static government page	Sitemap + targeted parser	Monthly
Manual source	Curator entry	On change

22. Admin and curation console

- Source Registry with enable/disable, schedule, rate limit and last run.
- Harvest run log with added/changed/failed counts.
- Duplicate candidate review.
- Metadata conflict diff view.
- Licence uncertainty queue.
- Broken-link queue.
- Community submissions moderation.
- Dataset merge and supersession controls.
- Publisher verification.
- Research Kit editor.
- Compatibility-rule editor with test fixtures.
- Audit log and rollback for metadata edits.

23. Security, privacy and abuse prevention

- Anonymous catalogue browsing.
- Account required only for synced projects/submissions.
- Minimal personal data.
- CSRF/XSS/SQL injection protections and strict server-side validation.
- URL allow/deny logic to prevent SSRF in link-check and submission pipelines.
- Crawler egress rules.
- Rate-limit search and submission endpoints.
- CAPTCHA or proof-of-work only if abuse appears; do not burden normal research use.
- Sanitise imported HTML metadata.
- Store secrets in environment/secret manager.
- Backups, migrations and restore testing.
- Private user-added dataset metadata stays private unless explicitly published.

24. Product analytics - privacy-respecting

- Search queries with anonymised aggregation.
- Zero-result searches.
- Dataset detail views.
- Add-to-Basket rate.
- Project completion rate.
- Report export rate.
- Broken-link reports.
- Most used Research Kits.
- Dataset source reliability.
- Crawler freshness and error rates.

25. Non-functional requirements

Requirement	Target
Performance	Search results < 500 ms server response for common queries; page interactive < 2.5 s on typical broadband.
Availability	99.5% MVP target; public catalogue should fail gracefully if background harvesters are down.
SEO	Dataset and publisher pages server-rendered with schema.org Dataset markup where appropriate.
Scalability	Design for 100k-1m metadata records without changing the canonical model.
Accessibility	WCAG 2.2 AA.
Internationalisation	Schema supports countries, languages and regional date/number formats even if UI launches in English.
Open source	Reproducible local setup, seed fixtures, migrations and contributor documentation.
Reproducibility	Exports contain dataset IDs, source URLs, versions and timestamps.

26. MVP definition

26.1 Must ship

- Home + Explore + Dataset Detail.
- Search and facets.
- At least 500-2,000 high-value seeded dataset records in the launch domain.
- Project Basket.
- Project Workspace.
- Basic compatibility rules.
- PDF/HTML/CSV/JSON Research Data Plan export.
- At least 5 Research Kits.
- Source Registry + 3-5 structured harvest adapters.
- Admin curation queue.
- Licence display and source-link policy.
- Anonymous use + optional accounts.

- Public documentation of ranking, metadata and licensing model.

26.2 Explicitly not MVP

- Generative AI.
- Automated research conclusions.
- Hosting large third-party datasets.
- Social feed or follower system.
- Full collaborative editing.
- Complex recommendation ML.
- Notebook execution.
- Paid marketplace.

27. Roadmap

Phase	Goal	Key additions
Phase 0 - Foundation	Validate concept	Schema, 100 manually curated records, clickable prototype, 2 Research Kits.
Phase 1 - MVP	Useful standalone product	Catalogue, search, basket, projects, rule engine, reports, initial harvesters.
Phase 2 - Trust & scale	Broader adoption	Publisher pages, submissions, better geospatial search, saved searches, dataset change tracking.
Phase 3 - Collaboration	Research team workflows	Shared projects, comments, public project templates, DOI-style project snapshots if appropriate.
Phase 4 - Optional intelligence	AI-enhanced but not AI-dependent	Natural-language query parsing, metadata summarisation, research question to kit suggestions, gap explanation.
Phase 5 - Platform	Research ecosystem	Public API, integrations with Zotero/Notion/GitHub/Jupyter, plugins, institution deployments.

28. Future AI layer - optional and replaceable

AI should be an enhancement, never a structural dependency. All AI outputs must be traceable to indexed metadata and must not silently invent dataset properties.

- Natural-language research question -> structured filters.
- Suggest Research Kits and dataset roles.
- Summarise long documentation pages.
- Explain compatibility warnings in plain English.
- Draft a data-acquisition methods paragraph using only selected metadata.
- Suggest alternative datasets when a selected dataset is unavailable.
- Never change source licence or factual metadata without evidence.

29. Testing and quality assurance

- Unit tests for every compatibility rule.
- Contract tests for each harvester adapter.
- Golden metadata fixtures from real portals.
- Duplicate-detection tests.
- Link checker mocks and timeout tests.
- Licence mapping tests.
- Search relevance evaluation set.
- Accessibility automated + manual tests.
- Export snapshot tests.
- Security tests for malicious URLs and imported HTML.
- End-to-end journeys: question -> search -> basket -> project -> report.

30. MVP acceptance criteria

68. A new user can find a dataset from the home page in fewer than three interactions.
69. A user can add at least 20 datasets to one project and assign roles.
70. The platform can explain at least geography, time, licence and identifier compatibility.
71. The generated report includes authoritative source links and current metadata snapshots.
72. No dataset download is proxied through the platform unless an explicit hosting rule allows it.
73. Crawler failures cannot corrupt published records.
74. Every published dataset has provenance and a last-checked timestamp.
75. A source record can be corrected, superseded or removed through admin workflow.
76. The entire MVP can run locally from documented open-source setup instructions without any paid AI API.

31. Worked use case - built environment

Research question

“How do urban morphology and local weather conditions influence summertime overheating risk in residential buildings across Leeds?”

User builds the project

Research role	Selected dataset type	What the platform contributes
Study boundary	Leeds administrative boundary	Official source link, geometry metadata, licence.
Building stock	EPC / building attributes	Coverage, years, fields, access restrictions, related identifier datasets.
Urban morphology	Building footprints / heights / LCZ	Spatial resolution, CRS, coverage and join notes.
Weather	Observed/gridded weather + simulation weather files	Temporal resolution, years, station/grid coverage.
Vulnerability	Census / deprivation / fuel poverty	Geography level and temporal mismatch warnings.
Reference	Official climate station / validation source	Role completeness and relationship to weather source.

The final report does not analyse overheating. It documents the evidence architecture needed to perform the study, identifies missing or mismatched components, and gives the researcher one reproducible place to return to every authoritative dataset.

32. Naming and brand direction

The long-term brand should not be constrained to architecture or built-environment terminology. The product concept is closer to “navigation through research evidence and datasets”. A final name should be trademark/domain checked before adoption.

- Brand territory: curiosity, navigation, evidence, discovery, pathways, research infrastructure.
- Avoid names that imply the platform owns or hosts all data.
- Avoid “AI” in the name because the core product is intentionally useful without AI.
- A descriptive subtitle can carry meaning even if the master brand is invented, e.g. “[Brand] - Research Data Navigator”.

33. Recommended implementation order

77. Canonical metadata schema + migrations.
78. Seed script with 100 curated records.
79. Dataset catalogue and detail pages.
80. Search + filters.
81. Basket state.
82. Project model and workspace.
83. Compatibility engine with 4 core rule families.
84. Report data model + HTML export, then PDF.
85. Research Kits.
86. Admin console.
87. Structured harvesters.
88. Community submissions.
89. Scale search and map features only when data volume warrants it.

34. Suggested repository structure

```
apps/
  web/              # Next.js UI
  api/              # FastAPI application
workers/
  harvester/        # source adapters + schedules
  link_checker/
packages/
  schema/           # canonical metadata models
  compatibility/    # deterministic rules
  reporting/        # HTML/PDF/CSV/JSON exports
  vocabularies/     # licence, format, geography mappings
  ui/               # shared components
data/
```

```
  fixtures/
  seeds/
  research_kits/
infra/
  docker/
  migrations/
  deployment/
docs/
  methodology/
  source_adapter_guide/
  contributor_guide/
  api/
tests/
  unit/
  integration/
  e2e/
```

35. Master build prompt for an autonomous coding agent

> **Instruction**
> Use the following as the root implementation brief. The agent should build in vertical slices, keep the app runnable after every milestone, and never invent external dataset licences or metadata.

Build an open-source web application called “Research Navigator” (working title) according to this specification. Its core purpose is to index authoritative dataset metadata, help a researcher discover and compare datasets, add them to a Project Basket, organise them into a research project, evaluate deterministic compatibility rules, and export a citable Research Data Plan.

Hard constraints:
1. V1 must work without any generative-AI API.
2. Do not mirror third-party dataset content by default. Store metadata and official source/access/documentation links.
3. Use PostgreSQL + PostGIS as the canonical database.
4. Implement a typed canonical metadata schema and migrations before crawler logic.
5. Build source adapters for structured catalogues before HTML scraping.
6. Every dataset must retain provenance, source record ID where available, last-checked timestamp, licence state and source URL.
7. Compatibility checks must be deterministic and explainable.
8. The report generator must export HTML, PDF, CSV and JSON.
9. Keep anonymous browsing open; accounts are only required for synced/saved projects and submissions.
10. Target WCAG 2.2 AA and responsive design.

Start with the built-environment seed domain but ensure the taxonomy and schema are domain-neutral. Implement in this order: schema -> seed data -> catalogue -> detail page -> search/facets -> basket -> projects -> compatibility -> reports -> research kits -> admin -> harvesters -> submissions. Add automated tests and documentation at each stage. Do not add features outside the spec until MVP acceptance criteria pass.

36. Final product vision

The strongest version of this product is not “Kaggle without hosting”. It is the place where a research project acquires its data architecture. The catalogue gets the researcher in the door; the Project Basket, compatibility model, Research Kits and Research Data Plan are what make the platform habit-forming. If built carefully, a user should be able to begin with a vague question and leave with a transparent, citable, legally safer and practically executable dataset strategy - while every source remains connected to its original publisher.

North-star metric
Number of research projects that reach a complete, exported dataset plan - not raw page views or number of indexed datasets.